

Research article

The interactive effects of individual boundary spanning and boundary buffering when using enterprise social media

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ABSTRACT

Enterprise social media (ESM) provides convenient channels for employees to access enterprise-wide work resources but also exposes employees to potential distractions. Prior studies have confirmed this “two-sided” role of ESM usage on employee work performance. Ongoing studies are exploring possible methods to alleviate side effects of ESM usage. Drawing on boundary management literature and information processing theory, we propose the concept of individual boundary spanning to capture employees’ information-obtaining behaviors via ESM and identify two mediating mechanisms—mindfulness and information overload—through which an individual’s boundary spanning affects work performance. We also propose the concept of individual boundary buffering to capture protective actions by individuals that prevent them from becoming disturbed or overloaded with ESM requests. We further investigate the interaction effects of these two boundary activities in the ESM context. We conducted a survey and collected data from 202 ESM users from a research and development enterprise in China. The results suggest that individual boundary spanning triggers mindfulness in individuals that benefits their work performance but also increases information overload, harming work performance. Individual boundary buffering can enhance the positive relationship between individual boundary spanning and mindfulness, and mitigate the positive relationship between spanning activity and information overload.

1. Introduction

Companies are increasingly deploying enterprise social media (ESM) as internal information and communication tools (ICTs) and resource-sharing platforms to support employees’ work (Cai, Huang, Liu, & Wang, 2018; Leonardi, 2014; Shang & Sun, 2020). However, recent studies have shown that organizations have vague guidelines regarding the use of these tools that enable employees’ work to benefit from enterprise-wide resources (Lu & Pan, 2019). One investigation showed that roughly half of the companies surveyed could not effectively use ESM tools (Li, 2015). Since social media has been considered a significant source of distraction and waste of work resources, managers are still uncertain as to whether ESM is a “productivity killer” (Archer-Brown & Kietzmann, 2018). Therefore, how organizations can better leverage the advantages of ESM and alleviate the potentially adverse effects of ESM use has become an important and urgent issue (Liang, Xin, Yan, & Jianxiang, 2020).

One main advantage of ESM is that it offers unprecedented

opportunities for employees to break the work boundary formed by work or department divisions, enabling them to conveniently access and share information-based resources across the company (Ellison, Gibbs, & Weber, 2015; Leonardi, Huysman, & Steinfield, 2013). According to boundary management theory, behaviors that solicit external resources through action and communication are recognized as boundary spanning (Dey & Ganesh, 2017; Tushman & Scanlan, 1981; Yan & Louis, 1999).

Scholars have provided evidence that ESM can support boundary spanning within organizations. For example, Pee (2018) found ESM’s various affordances (e.g., visibility) can alleviate codification efforts for cross-domain or complex knowledge transmission. Rode (2016) and Liu and Bakici (2019) recognized ESM as a valuable tool to help individuals search for work-related information. Aboelmaged (2018) found that employees can acquire knowledge from individuals both within and outside organizations via ESM. Scholars posit that work completion generally requires employees to have access to adequate work resources (Bacharach & Bamberger, 1995; Tatikonda & Rosenthal, 2000),

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therefore, boundary spanning can benefit individuals' work by complementing knowledge and expertise, etc. Nevertheless, some ESM scholars have observed that even though ESM provides convenient information channels for both work and non-work-related communications, the abundance of information may render users in a work environment overwhelmed, consuming their limited work resources and ultimately leading to information overload (Chen & Wei, 2019; Zhang, Zhao, Lu, & Yang, 2016). These studies have provided valuable insights into our understanding of the positive and negative effects of ESM usage. Unfortunately, how organizations can alleviate the potential negative effects of ESM use on individuals and improve employees' usage of ESM has still not been answered.

Scholars have suggested that individuals take protective actions such as neglect and avoidance behaviors to protect themselves from information overload and prevent interruption from computers and communication tools during work time (Russell, Jackson, & Banks, 2019; Zhang Y, 2020). According to boundary management theory, behaviors individuals take to protect themselves from becoming disturbed and overloaded from the outside can be defined as "boundary buffering" (Faraj & Yan, 2009). However, when researching boundary spanning and boundary buffering, extant studies mainly focus on the effects of these two kinds of activities individually (e.g., Faraj & Yan, 2009) or only consider one of them (e.g., Schotter, Mudambi, Doz, & Gaur, 2017; Van Osch & Steinfeld, 2018). There is currently a lack of empirical investigation into the interaction effects between these two types of boundary activities at the individual level. In their review of team boundary management, Dey and Ganesh recommended that scholars pay attention to the interaction effects of both boundary spanning and boundary buffering. They suggested that "buffering provides teams with slack, which gives team members an opportunity for quite reflection within the team. This fuels learning and innovation in the team by letting team members synthesize the knowledge they have gathered through boundary-spanning activities" (Dey & Ganesh, 2017).

This study argues that individual boundary spanning and boundary buffering can work together to achieve interaction effects for individuals' work for several reasons. First, within the ESM context, the objects of boundary spanning and boundary buffering are both information-type content (i.e., spanning for needed work-related information and buffering for non-work-related information), the two activities are compatible while being implemented. Second, ESM activities are information-based. Since individuals possess limited information-processing resources, the effectiveness of boundary spanning depends on whether individuals have adequate information processing resources (e.g., cognition) to process the work materials they acquire from spanning and processing efficiency. Third, individual boundary buffering can be practiced to ensure continuity of ongoing information processing procedures and save resources from unnecessary activities, which acts as an effective approach to strengthen the effectiveness of boundary spanning (Dey & Ganesh, 2017). This study also resonates with the call for the need to provide empirical studies to examine the effectiveness of protective actions for coping with information overload (Saxena & Lamest, 2018). We therefore propose our research question as: *What role does ESM boundary buffering play when considering the effects of ESM boundary spanning on individual work performance?*

We apply information processing theory as the theoretical lens through which to understand how humans process information to answer research questions (Miller, 1956). The theory suggests that human brains receive information from various sources and store it in working memory for further cognitive processing (Atkinson & Shiffrin, 1968; Simon, 1978). The information processing yielding two outcomes: mindfulness or information overload. Mindfulness is a mind mode in which individuals are open to new experiences and display active awareness of novel distinctions. In this mode, individuals tend to reflect on their situations, consider actions from various perspectives, and think of different possibilities, etc. (Frese & Zapf, 1994a; Thatcher, Wright, Sun, Zagenczyk, & Klein, 2018; Wei, Zhu, & Chen, 2020). However,

because humans have limited information processing capacity (e.g., working memory) (Miller, 1956), individuals who are continually accessing information will quickly reach their limits, leading to information overload (Swar, Hameed, & Reychav, 2017). This study considers mindfulness and information overload as two mediators that mediate the impact of individual boundary spanning on individual work performance.

Furthermore, we consider individual boundary buffering as a mechanism to strengthen individuals' information processing efficiency by avoiding unnecessary information processing and enhancing the levels of mindfulness. Based on a survey of an important research and development company in China, the results support the idea that ESM enabled individual boundary spanning to promote users' work performance by motivating mindfulness, but it also harmed work performance by increasing information overload. The results also support that properly utilizing individual boundary buffering can help employees enhance mindfulness and mitigate information overload. These findings significantly contribute to our understanding of ESM usage in several ways. They suggest individual boundary buffering, in addition to individual boundary spanning, as another necessary activity in effective ESM usage in the ESM literature. They confirm that information overload and mindfulness are two critical explanatory factors of the relationship between ESM-enabled individual boundary spanning and individual work performance. Finally, they provide new insight into the boundary management literature by revealing the interaction effects of boundary spanning and boundary buffering in the ESM context.

The remainder of the paper proceeds as follows. Section 2 summarizes prior literature on individual boundary spanning and individual boundary buffering. Section 3 introduces the theoretical background and hypothesis. Section 4 elaborates on our research method. Section 5 presents the results of the data analysis. Section 6 discusses theoretical and practical implications of the findings, and specifies the study's limitations. Section 7 concludes this study.

2. Literature review

Boundary management literature recognizes a series of activities in which individuals interact with the external environment, such as when communicating with external actors to obtain resources beyond their boundaries or maintaining boundaries to protect themselves from being negatively influenced by the outside environment (Dey & Ganesh, 2017; Wu, Huang, Jiang, Klein, & Liu, 2020). The scope of boundaries can be different when considering diverse analytical units and research contexts. For example, when the boundary is at an organizational level, the organization's scope can be classified as within or outside the company (Schotter et al., 2017). In the context of this study, ESM is typically implemented within the company and in the company we surveyed, project teams generally carry out the job responsibilities. Therefore, we define the scope of individual boundaries as that formed by the accessibility of resources the project team officially provides. For example, suppose one developer engages in a project with three other teammates. In that case, resources provided by the members within the project team are resources within the work boundary. At the same time, the knowledge of colleagues in the same department/organization who are not members of the project is a resource outside the work boundary.

2.1. Individual boundary spanning

In boundary management literature, individual boundary spanning refers to individual behaviors to solicit external work resources through actions and communication relationships (Tushman & Scanlan, 1981). Boundary spanning can play an important role in employees' work performance as it supplements their work resources (Bacharach & Bamberger, 1995; Tatikonda & Rosenthal, 2000; Wang et al., 2019). In an organization, coworkers are reliable sources for supplementing work resources, and prior research has found that employees' sharing

activities such as discussing ideas, updating information, and exchanging knowledge, can broaden the resources employees possess, improve the details of their work as well as their decision-making capabilities (Aboelmaged, 2018; Anand & Singh, 2011; Mäntymäki & Riemer, 2016; Teigland & Wasko, 2003). ESM has been suggested as effective tools for transferring information-based work resources among employees within organizations (Aboelmaged, 2018; Laitinen & Sivunen, 2020).

In particular, ESM connects every employee and equips multiple sharing and communication functions to support users in breaking boundaries established by structures such as work divisions, departments, or individuals' social networks so that they can access enterprise-wide resources (Leonardi et al., 2013; Van Osch & Steinfield, 2016). Prior studies have implicitly suggested that ESM supports individuals across work boundaries to acquire external resources to aid their work.

According to our summary, as shown in Table 1, studies that embody the connotation of individual boundary spanning via ESM can be divided into two streams: the knowledge management stream and the information management stream. In the former, scholars found that ESM can facilitate employees' knowledge sharing and knowledge creation. Pee (2018) found that ESM's various affordances alleviate the difficulties of sharing knowledge with others. Qi and Chau (2017) found that an employee's ESM usage is positively related to his/her knowledge sharing and knowledge creation, which benefits organizational learning. In regard to information management, scholars mainly focus on the antecedents of ESM information sharing (Laitinen & Sivunen, 2020) and information searching (Van Osch & Steinfield, 2016, 2018), and only one study has explored the relationship between information management and individual work performance (Lu & Pan, 2019).

However, there are significant differences between the above perspectives and the distinct boundary spanning perspective. First, extant studies do not explicitly discriminate whether knowledge or information management activities occurred within the work boundary (e.g., within the project team) or outside the work boundary but within the organization. Such lack of discrimination fails to recognize the benefits of ESM's broadening individuals' access to work resources across work boundaries. Second, neither the knowledge management perspective nor the information management perspective represent the complete range of individual boundary spanning activities. Presently, we only vaguely understand how ESM users can benefit from accessing the broader range of enterprise work resources outside their conventional work jurisdiction. We argue that boundary management theory provides a research lens to capture such behavior and investigate different mechanisms influencing an individual's work.

Besides the topic of ESM, many studies have investigated the influence of individual boundary spanning activities on other issues, such as the relationship between boundary spanning and managers' influence (Manev & Stevenson, 2001) or boundary spanning in multinational corporations (Barner-Rasmussen, Ehrnrooth, Koveshnikov, & Mäkelä,

2014; Reiche, 2011; Schotter et al., 2017). These studies found that individual boundary spanning can broaden the channels for acquiring work resources (Barner-Rasmussen et al., 2014; Reiche, 2011) while providing extra cues for decision-making (Floyd & Wooldridge, 1997), thus benefiting work performance. However, when comparing ESM-facilitated boundary spanning to these conventional individual boundary spanning studies, several differences remain. First, they examined different actors. ESM allows every employee to access the enterprise resource pool, thus everyone can be a boundary spanner (Schotter et al., 2017). However, conventional boundary spanning literature has shown that only a limited number of individuals possess key resources and can mediate activities across the boundary as boundary spanners. For example, inpatriates who can connect their home unit and the headquarters (Reiche, 2011), or IT employees who have unique knowledge and serve multiple departments (Wang et al., 2019) can be boundary spanners. Second, the resources boundary spanners can access are different. Every spanner can access the enterprise-wide resources pool in the ESM context, such as project reports and Q&As on the ESM. If spanners find it necessary, they can contact relevant colleagues for help through the tool even if they do not know one another. However, their work and relational network limit the resource pool of conventional boundary spanners. Lastly, motivations or expectations under spanning activities might differ since ESM users engage in boundary spanning activities mainly out of their own interests (i.e., acquiring support for their work). However, because of their unique identities (roles), conventional boundary spanners might be expected to contribute benefits beyond the individual level, such as promoting knowledge transfer between the home unit and headquarters (Reiche, 2011) or negotiating with external stakeholders on behalf of the team (Faraj & Yan, 2009). Inappropriate management of this conventional boundary spanning could lead to side effects such as role overload (Wang et al., 2019). As ESM becomes more prevalent in businesses, these differences indicate that ESM boundary spanning is a topic that warrants further investigation. In this study, we investigate the mechanisms through which ESM-based individual boundary spanning could affect individuals' work performance.

2.2. Individual boundary buffering

Boundary buffering generally refers to behaviors that shield the entity from unwanted exposure to the environment (Faraj & Yan, 2009). In the ESM context, individual boundary buffering can be defined as actions to insulate individuals from unwanted interference and overloaded demands from the ESM. In contrast to boundary spanning, which can be characterized as a form of engagement activity, boundary buffering is a form of disengagement (Faraj & Yan, 2009). Boundary buffering decreases outside influences, strengthening the effectiveness of rational action inside (Scott, 1998).

Researchers have thus far paid attention to team-level boundary buffering (Dey & Ganesh, 2017). For example, Chang, Wu, Discenza, and Klein (2021) found that team boundary buffering can improve software development project performance. Faraj and Yan (2009) found that team boundary buffering can enhance team members' psychological safety and team performance. Wu et al. (2020) found that supplier teams' boundary buffering in the supply chain management context can protect teams from suffering the environmental uncertainties driven by buyers' requirements changes. Team boundary buffering activities are often carried out by team leaders dedicated to creating a well-protected working environment. In the ESM context, ESM makes it viable and convenient for individuals to practice individual-level boundary buffering. Except for conventional buffering approaches such as negotiation or coordination (Faraj & Yan, 2009), the use of ESM functionalities can be regarded as a new, technology-based method of buffering, which is complementary to extant buffering approaches. Specifically, conventional buffering activities generally require interaction engagement, such as refusing a non-work invitation during work time. An individual

Table 1
The summary of boundary spanning in the ESM context.

Boundary spanning in the ESM context	Topic-related works
Knowledge management	• Knowledge sharing (Pee, 2018; Qi & Chau, 2017; Rode, 2016; Kwahk & Park, 2016; Sun et al., 2019; Filstad et al., 2018; Oostervink et al., 2016; Razmerita et al., 2016; Fulk & Yuan, 2013) • Knowledge creation (Qi & Chau, 2017; Cao et al., 2020; Chen et al., 2020)
Information management	• Information sharing (Laitinen & Sivunen, 2020; Lu & Pan, 2019) • Information seeking/searching (Liu & Bakici, 2019; Lu & Pan, 2019) • Team boundary spanning-information searching (Van Osch & Steinfield, 2016, 2018)

can decide to reject or not only after processing the relevant information, but such information processing action will have already interrupted current work. However, ESM functionalities such as blocking certain information sources, using mute mode (i.e., muting the notification of all or specific kinds of messages), or online status settings (i.e., setting online status to busy or in a meeting when one doesn't want to be disturbed) can achieve the buffering effect without interrupting employees' current activities. For knowledge workers, guaranteeing the continuity of their intellectual process often means guarding high-performance work (Jett & George, 2003).

These technological methods of boundary buffering also echo prior studies' suggested approaches to coping with information overload in the contemporary era of information explosion, such as queuing (i.e., postponing information processing to less busy times) (Wilson, 1995), filtering (i.e., weeding out useless information) and withdrawing (i.e., limiting the volume of information that one is exposed to) (Savolainen, 2007). Nevertheless, scholars have pointed out a lack of empirical studies to statistically investigate the effectiveness of approaches for coping with information overload (Saxena & Lamest, 2018). Since contextualization is essential to improving the theory's generalizability (Saxena & Lamest, 2018), this study aims to investigate the influence of individual boundary buffering in the ESM context.

In short, boundary management studies have suggested that different boundary activities have different motivations, targets and objects, and functions (Dey & Ganesh, 2017; Marrone, 2010). We summarized the properties of individual boundary spanning and buffering in the ESM context in Table 2.

Different motivations drive ESM users to practice different boundary activities. Specifically, individuals who practice boundary spanning aim to obtain more resources that once were difficult or unable to be accessed (Leonardi et al., 2013). Individuals engage in boundary buffering because they want to remain undisturbed and not be overloaded with ESM requests (Faraj & Yan, 2009). Targets of boundary spanning are external colleagues who possess information, knowledge, or technology that spanners need to aid them in finishing their work, and external work resources stored on ESM. The targets of boundary buffering are external ESM users who may interrupt individuals or give them an overload of requests during work time. Along with the different boundary targets, boundary objects for boundary spanning are task-related documents, knowledge, ideas, feedback, and conversations enabled by ESM. Simultaneously, non-work-related posts, notifications, chats, or overloaded work requests are objects that ESM users buffer during work time (Leonardi et al., 2013).

Table 2
Properties of individual boundary spanning and buffering in the ESM context.

Definition and scope of boundary: We follow the general definition of boundary given by Yan and Louis (1999) "domain of interactions of a system with its environment to maintain the system as a system and provide for its long-term survival." We specify the scope of individual boundaries in the ESM context as formed by the accessibility officially provided by the project team.		
Properties	Individual boundary spanning	Individual boundary buffering
Boundary motivation	Use ESM to obtain resources external to one's work boundary.	Protect oneself from disturbances or request overload on ESM.
Boundary targets	External ESM users (colleagues) who grasp the information, the knowledge or technology that one needs to finish the work, and external work resources stored on ESM.	External ESM users who may interrupt the individual or overload them with individual requests during work time.
Boundary objects	Task/project-related documents, knowledge, ideas, feedback, conversations enabled by ESM.	Non-work-related posts, notifications, chats, or overloaded work-related requests on ESM.

3. Theoretical background and hypotheses development

3.1. Information processing theory

Given that information processing theory (IPT) has been proposed as one important theoretical lens through which to explore the influence of boundary spanning on individuals' work (Carlile, 2004; Wang et al., 2019), our study adopts it to formulate the hypothesized relationships. Information processing theory provides scholars with a theoretical lens to understand how humans process information (Atkinson & Shiffrin, 1968; Miller, 1956). According to theory, the human brain receives information from various sources and stores it in working or short-term memory. There, it is subject to further cognitive processing (Atkinson & Shiffrin, 1968; Simon, 1978). Prior studies have suggested that individuals can engage in two cognitive processing states to process the information stored in working memory: mindfulness and mindlessness (Langer, 1989, 1992). Mindfulness is defined as a state of cognitive functioning in a specific situation under which individuals exhibit conscious awareness and alertness of their actions and contexts, openness to new experiences (Addas & Pinsonneault, 2018; Langer, 1989), a tendency to consider actions from various perspectives, reflect on their situations, think of different possibilities, etc. (Frese & Zapf, 1994a; Hacker, 2003). When individuals are in a mindless state, their thoughts, emotions, and behaviors rely on well-structured routines based on distinctions and associations learned in the past (Bodner & Langer, 2001; Haigh, Moore, Kashdan, & Fresco, 2011). Mindlessness requires less effort, but people are also more likely to neglect the nuances of new information in such a processing mode. Scholars posit that individual work performance is determined by whether employees can access adequate work resources such as information, knowledge and expertise (Bacharach & Bamberger, 1995; Tatikonda & Rosenthal, 2000), which stresses the importance of how individuals process the information-based work resources.

3.1.1. Theorizing the relationship between boundary spanning and mindfulness

Scholars have adopted the information processing perspective to investigate the influence of boundary spanning activities on boundary spanners (Carlile, 2004; Wang et al., 2019). They proposed that the objects of boundary spanning activities are generally information flows, which convey work-related information and knowledge. ESM users consciously span boundaries to acquire task-related information that usually contains information outside the spanner's routine experience that could update or complement the spanner's understanding of the work. Under such situation, mindfulness is validated as an appropriate information processing mode that allows individuals to effectively leverage the information obtained to benefit their current work (Addas & Pinsonneault, 2018; Jett & George, 2003). Therefore, we use mindfulness to capture the potential positive influence of boundary spanning activities on an individual's work.

3.1.2. Theorizing the relationship between boundary spanning and information overload

Information processing theory acknowledges individuals possess limited information processing capacity (Gao, Liu, Guo, & Li, 2018; Miller, 1956). Considering boundary spanning brings in external work-related information and knowledge for spanners (Ratcheva, 2009; Teigland & Wasko, 2003; Wang et al., 2019), reflecting the need for information processing, we argue that intensive boundary spanning activities will import too much information to spanners' working memory over a short period, causing an individual's information processing capacity to quickly reach its limits (Swar et al., 2017) and resulting in information overload (Gao et al., 2018; Jacoby, Speller, & Kohn, 1974).

3.1.3. Theorizing the interaction between boundary spanning and buffering

Information processing theory emphasizes that the fit between the quantity of information to be processed and the capacity to process it will determine individual performance, such as decision making quality (Gao, Zhang, Wang, & Ba, 2012). We first map the boundary spanning to determine the quantity of information to be processed. Further, boundary buffering can shield individuals from unwanted interference and overloaded demands (Wu et al., 2020), maintaining their information processing capacity and promoting information processing efficiency (Dey & Ganesh, 2017). Therefore, information processing theory supports our argument for positive interaction effects between individual boundary spanning and boundary buffering.

Fig. 1 shows how a boundary spanner exploits information from ESM.

3.2. Hypotheses development

By drawing on information processing theory and the relevant literature, this study proposes a research model in which individual boundary spanning is positively related to mindfulness and overload. Mindfulness could promote individual performance, but overload would decrease individual performance. We further investigate the moderating effects of individual boundary buffering, which is hypothesized to strengthen the positive effect of boundary spanning on mindfulness while mitigating the positive effect of boundary spanning on overload. Fig. 2 presents the research model. The hypotheses are developed in the paragraphs that follow.

As discussed, boundary spanning is a conscious activity that aims to acquire useful resources to aid spanners' work, such as finding novel task solutions or filling knowledge gaps to develop individuals' understanding of their work. So individuals have the motivation and willingness to keep an eye on the link between external resources and their current work. In the ESM context, resources obtained from boundary spanning are information-based, which requires effective information processing procedures. This specific requirement aligns with the practice of spanners' mindfulness because individuals in the mindful state are alert, lively, aware of current experiences (Baer, 2003; Roberts, Thatcher, & Klein, 2007), more willing to explore alternatives and novelties brought by external resources (Frese & Zapf, 1994b; Hacker, 2003) and more sensitized to the subtleties of context (Roberts et al., 2007). This mindful state enables them to capture the similarities and differences between external resources and current work and identify the formation factors behind them. Therefore, mindfulness is a processing mode that can help individuals fulfill the goal of boundary spanning.

Prior research suggests that task-related events can trigger an individual's mindful processing mode (Addas & Pinsonneault, 2018; Jett & George, 2003; Langer, 1989). In the ESM context, external resources acquired from boundary spanning via ESM may contain novelties or discrepancies about spanners' work, serving as external clues to trigger individuals' mindful processing to help them better integrate external resources. Hence, we posit the following hypothesis:

H1a. Individual boundary spanning is positively related to mindfulness.

ESM activities are information-based. Individuals engaging in boundary spanning activities generally need to go through a series of information processing processes, including collecting potentially useful resources (either in proactive ways such as searching on the ESM or in passive ways such as receiving relevant feedback) and recognizing whether these resources match their needs. If the resource fits their needs, substantial cognitive efforts must be made to incorporate such external resources to aid the spanner's work (Addas & Pinsonneault, 2018; Carlile, 2004). The more boundary spanning activities there are, the more information processing capabilities and resources are required (Swar et al., 2017). However, information processing theory and related studies have proposed that individuals have limited work resources (Miller, 1956), and intensive boundary spanning activities could drive information processing resource deficiency and lead to information overload (Cao & Yu, 2019; Sun et al., 2021).

On the other hand, individuals face various work-related activities aside from boundary spanning via ESM during work time. Then when an individual's work resources are rapidly depleting due to boundary spanning, few resources are left to process the other necessary tasks. Therefore, an incompatibility between the limited resources one possesses and the information processing required is aggravated, leading to information overload (Gao et al., 2018). Hence, we propose the following hypothesis:

H1b. Individual boundary spanning is positively related to information overload.

Through mindful processing, an individual can incorporate relevant external resources gathered from ESM into contextual-fit internal resources to assist their work (Addas & Pinsonneault, 2018). For example, a developer can search a particular development flow from a completed project report (e.g., similar to the one he is currently working in) uploaded by other colleagues on the ESM. Even though two projects may have different aims, objects, and personnel, some test procedures can be referenced, and the developer can compare and analyze the differences and commonalities between the two projects and recognize and acquire the key knowledge to benefit his work when in a state of mindfulness.

The mindful process can also improve individuals' work details (Addas & Pinsonneault, 2018). In this mindful state, individuals are sensitive to discrepancies and abnormalities in their work (Frese & Zapf, 1994a) and possess the motivation to improve their performance (Roe, 1999). Therefore, they are efficient and effective in intellectual endeavors to solve their problems and are open to changing routinized behaviors such as altering existing plans or developing new approaches based on the current situation, correcting their work process, or applying new approaches to work. Hence, we posit the following hypothesis:

H2. Mindfulness is positively related to individual work performance.

Information overload could decrease individuals' work performance. Bounded rationality emphasizes that individuals' cognitive limitations inhibit their decision-making (Karr-Wisniewski & Lu, 2010). Being overwhelmed by ESM information could render users unable to effectively recognize, interpret and integrate external resources, resulting in decreased decision-making efficiency (Jackson & Farzaneh, 2012; Keller

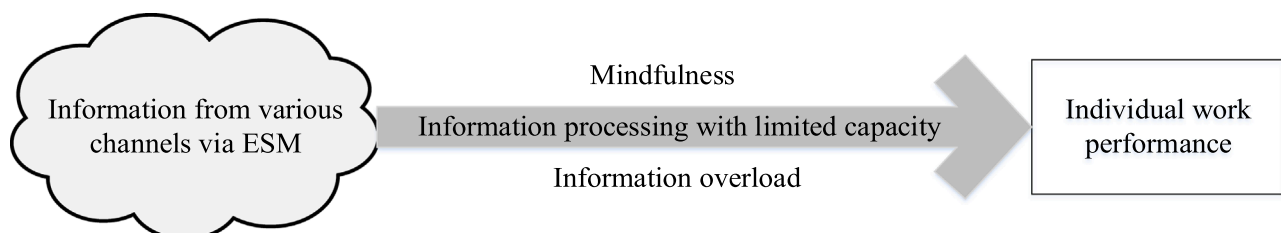


Fig. 1. Information processing model of ESM boundary spanning.

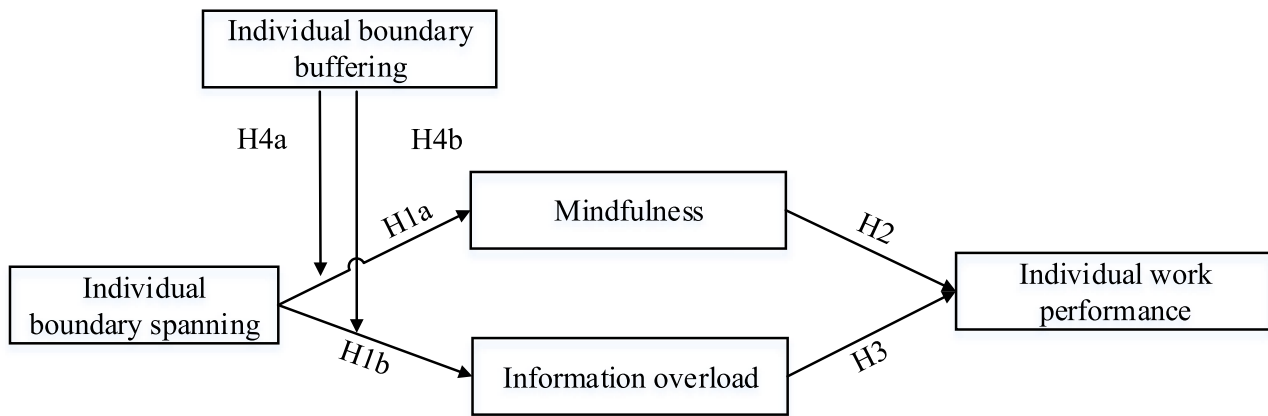


Fig. 2. Research model and hypotheses.

& Staelin, 1987; Schick, Gordon, & Haka, 1990) and increased errors (Eyrolle & Cellier, 2000).

In addition, when workers' minds are occupied with a large amount of ESM information, especially non-work-related information, it becomes harder for them to recall prior information and resume interrupted activities (Baethge & Rigotti, 2013; Schick et al., 1990). This predicament requires them to consume more work resources during the transition process and generate issues due to resource deficiency, such as work delays and decreased work performance (Adler & Benbunan-Fich, 2012). Hence, we posit the following:

H3. Information overload is negatively related to individual work performance.

3.2.1. Moderating effects

As Table 2 shows, both boundary spanning and boundary buffering objects are information-based content. The core rationale of their interaction is that individuals possess limited information processing resources (Miller, 1956). Specifically, on the one hand, achieving the benefits of boundary spanning requires individuals to process external information-based work materials obtained from spanning activities effectively (Carlile, 2004). Therefore, the more information processing resources (e.g., attention, time cognitive effort, etc.) available, and the higher the resource efficiency, the better the effectiveness of the boundary spanning. On the other hand, ESM boundary buffering can protect individuals from being exposed to an interruptive environment during work time and spending information processing resources on unnecessary ESM activities (Faraj & Yan, 2009; Galluch, Grover, & Thatcher, 2015). Boundary buffering guarantees the continuous running of information processing procedures and saves individuals' limited information processing resources. Therefore, it is reasonable to infer that the information processing procedure dynamics driven by boundary buffering can affect the outcomes of boundary spanning. Below, we discuss the moderating effects of boundary buffering concerning the relationship between boundary spanning and two mediators.

Engaging in mindful processing requires an individual's sustained and focused attention and intellectual efforts to analyze and synthesize the information acquired from ESM boundary spanning with current work details (Bishop et al., 2004). For example, a product manager may need to consider the underlying factors that cause discrepancies in the product evaluations collected from various department colleagues on ESM to leverage its report and experience better to adjust the development plan. Boundary buffering activities support such engagement by protecting themselves against unrelated activities to form a less disturbed working environment. This behavior ultimately enhances the continuity of their mindful process and rational behaviors (Scott, 1998).

On the other hand, boundary buffering provides individuals with

more slack to fuel mindful processing of information from boundary spanning (Dey & Ganesh, 2017). Due to an individual's limited working memory capacity and other work resources (Miller, 1956), subsequent unnecessary resource consumption is avoided when individuals block unrelated activities' involvement during work time (Faraj & Yan, 2009). Therefore, users can invest more work resources in processing information gathered from boundary spanning activities in a mindful mode. Hence, we posit the following:

H4a. Individual boundary buffering can strengthen the positive relationship between individual boundary spanning and mindfulness.

The convenient communication feature of ESM exposes employees to a broader interaction environment, aggravating the risk of receiving an overload of requests either out of work or non-work purposes (Chen & Wei, 2019; Guo, Lu, Kuang, & Wang, 2020). Engaging in appropriate boundary buffering activities can effectively defend ESM users from getting too much information processing pressure by ESM requests in two ways. First, individuals can refuse unrelated or overloaded requests from others to reduce information processing needs and avoid unnecessary consumption of work resources (e.g., working memory) related to this information and follow-up activities (Dey & Ganesh, 2017; Faraj & Yan, 2009). Secondly, individuals can take measures to decrease the disturbances from ESM information (Russell et al., 2019; Zhang Y, 2020). For example, users can turn on the ESM silent mode or mute unrelated notifications to avoid interruption or be reminded to check new messages every time they are received. Instead, they can handle these unread messages when they are free. In addition, checking collected unread messages at fewer time points means fewer transition processes between current work and ESM activities (Galluch et al., 2015), which helps employees save their work resources (e.g., time, cognition efforts).

Since individuals have limited resources to process information from both boundary spanning and other activities, they practice boundary buffering to decrease the expenditure of resources on unnecessary ESM information processing, leading to a decrease in the perception of incompatibility between boundary spanning and information processing resources. Hence, we posit the following:

H4b. Individual boundary buffering can mitigate the positive relationship between individual boundary spanning and information overload.

4. Methodology

4.1. Construct measurement

Multi-item scales operationalized constructs in this study. Items were adapted from validated existing studies; we modified some items to

ensure that they fit our context. A seven-point Likert scale was used as an anchor to measure all items. Specifically, *individual boundary spanning* and *individual boundary buffering* are adapted from Faraj and Yan (2009). *Individual boundary spanning* is measured by four reflective items and *individual boundary buffering* is measured by three reflective items, both of them are adjusted to fit the context of an individual's ESM use. *Information overload* is defined as the condition induced by the large amount of information generated on ESM that exceeds the capacity an individual can process (Jacoby et al., 1974). Five reflective items of the construct were adapted from Zhang et al. (2016). *Mindfulness* is defined as a mode of cognitive functioning in a specific situation that increases individuals' alertness to discrepancies, openness to newness, orientation in the present, and awareness of multi-angle thinking while performing key activities (Addas & Pinsonneault, 2018). The construct was measured with six reflective items adopted from (Addas & Pinsonneault, 2018; Langer, 1989). *Individual work performance* was evaluated by the ESM user's section leader, and four items from Venkatesh, Rai, and Maruping (2018) were used to reflect an individual's work performance in a particular project.

Since the study was implemented in China, the existing construct items were first translated into Chinese by three researchers fluent in English and Chinese. After minor corrections, we conducted two independent card sorting analysis rounds with eight Ph.D. students for initial validity (Moore & Benbasat, 1991; Tsai, Wu, Klein, & Jiang, 2021; Zhang, Guo, Jiang, Wu, & Jiang, 2021). Four students classified items according to their definition similarity and descriptions of construct categories in each round. First round classification accuracy was 93%. After minor corrections based on feedback, modified items were classified by the other four students in a second round following which the accuracy of classifications increased to approximately 96%. Improvements indicated adequate face and convergent validity.

At the company surveyed in this study, work is generally carried out by project teams. One can legitimately access resources in the project team, freely contact teammates, and respond promptly to requests made to ensure the project's progression. Therefore, we specified the "external" (the connotation of boundary) of the respondents as the demarcation formed by the accessibility officially provided by the project team that they are in.

In addition, items for boundary management constructs were adapted from Faraj and Yan (2009). In their study, items were used to measure team-level boundary activities in a general context. Compared to this study, both the research context and research level are different. Therefore, we conducted interviews with three employees at the company we surveyed who were experienced in using ESM to refine the wording of these items to ensure they were consistent with ESM users' actual use scenario. After incorporating their comments, we made several necessary amendments to these items, including specifying that all boundary activities are enabled by ESM and practiced by ESM users themselves. The statement sentences were also revised to fit our research purpose. For example, one item of boundary spanning in Faraj's study is "To what extent does the team encourage its members to solicit information and resources from elsewhere in and/or beyond the division?" We revised this item as "I use ESM to solicit information and resources from external channels." To measure boundary buffering, we repeated these steps but added more materials (i.e., use of typical ESM functionalities, which could reflect the connotation of ESM boundary buffering) to some of the items to aid respondents' comprehension. For example, one item of boundary buffering in Faraj's study is "To what extent are outside pressures deflected or absorbed so that the team can work free of interference?" We revised the item and added relevant materials, to "I take measures (e.g., muting nonwork-related notifications) to ensure that I can work free of ESM interference."

To further ensure the questionnaire's content validity, we invited ten experienced employees using ESM to participate in the pretest. They reviewed every item to assess precision and clarity as well as appropriate meaning in the ESM context, rating each item as "1-not matched," "2-

somewhat matched," "3-quite matched" and "4-highly matched." We then calculated the content validity index for each item. The results show that the index for each item is no less than 0.9, indicating good content validity (Polit & Beck, 2006). Several items were slightly modified according to feedback. The results from the pretest suggested that the measurement items were valid.

4.1.1. Control variables

Our study included seven control variables. *Polychronicity orientation* refers to the extent to which one prefers to be engaged in two or more tasks or events simultaneously. Two items were adapted from A. Chen and Karahanna (2018). Prior studies have validated that people with different polychronicity orientations show variations in resource allocation and performance when they face multiple tasks (Bluedorn Allen, Kalliath Thomas, Strube Michael, & Martin Gregg, 1999; Cameron & Webster, 2012). Demographic factors, such as *age*, *gender*, and *education* were also controlled in this study since they have been found to influence workload perception and work performance in the information and communication tool context (Ragu-Nathan, Tarafdar, Ragu-Nathan, & Tu, 2008; Tams, Thatcher, & Grover, 2018). We include *work experience* as a control, because more experienced workers could have more social and work ties, which could further influence information overload and work performance (Peng, Heim, & Mallick, 2015). Moreover, *ESM use experience* is controlled given that individuals with more experience in ESM would manage their ESM use more appropriately and utilize ESM work resources better. Finally, we included *non-work ESM use* (e.g., social-related ESM use) as a control because extant studies have validated that such use can affect information overload (Chen & Wei, 2019) and further influence work performance (Kuegler, Smolnik, & Kane, 2015).

4.2. Data collection

We collected data from a large Chinese telecommunications and information technology corporation with over 10,000 employees in Xi'an, China. The firm uses an ESM application that runs on personal computers and mobile devices for internal communication and other employee activities. The software is similar to other popular enterprise social media platforms in China, such as DingDing and WeCom, which integrate various functions, including basic communication functions (e.g., instant messaging and online meetings), social features (e.g., posts) and work-support features (e.g., documentation management).

The formal survey used a multi-informant design to eliminate the risk of social desirability bias and common method variance. Section members are expected to answer most of the questionnaire, while section leaders are required to evaluate their own section members' performance. In addition, collecting all items in one survey wave could lead to common method bias (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). To help alleviate these issues, our study conducted a two-wave survey, which has been widely implemented (e.g., Wang, Wang, Lin, & Abdullat, 2021). Specifically, section members are required to answer their questionnaires over two waves. The first wave includes measurement of ESM boundary spanning and buffering, and polychronicity orientation. One week later, respondents finish the questionnaire, with measurements of mindfulness and information overload, and demographics.

With the help of the company's administrator, we finally sent 400 questionnaires and received 284 responses. After deleting responses with missing data, we analyzed 202 usable responses to test the proposed model. We compared the early and late respondents for all the constructs and found no significant differences between them, suggesting that nonresponse bias was not a problem (Armstrong & Overton, 1977). Table 3 shows respondents' demographic information.

Table 3
Sample Demographics.

Demographic Variables	Distribution
Age	≤ 25: 6%; 25–30: 32%; 31–35: 37%; 36–40: 21%; 41–45: 4%
Education	College graduate: 14%; Postgraduate degree: 84%; Ph.D degree: 2%
Gender	Male: 73%; Female: 27%
Work experience	≤ 5 years: 35%; 6–10 years: 30%; 11–15 years: 26%; 16–20 years: 8%; > 20 years: 1%
ESM experience	≤ 6 months: 12%; 7–12 months: 12%; 13–24months: 16%; > 24months: 60%
Numbers of other projects or tasks undergo currently	Null: 13%; One: 23%; Two: 33%; Three: 19%; Four: 5%; Five: 5%; ≥ Five: 2%

5. Data analysis and results

5.1. Measurement model

We used the partial least squares (PLS) method (SmartPLS 2.0) to test the measurement model and assess each construct's reliability and validity. PLS is best suited for our research because it does not require multivariate normality of the data (Teo, Wei, & Benbasat, 2003). Second, this method is suitable for explaining a relatively complex research model (Sarstedt, Ringle, & Hair, 2017). Third, PLS is capable of calculating reasonable latent variable scores of each construct. Rather than assume equal weights for all indicators of a scale, PLS allows every indicator to contribute differently to the composite score of the latent variable (Chin, Marcolin, & Newsted, 2003), which avoid fixed-scale construction (Lowry & Gaskin, 2014), the obtained latent variables support subsequent analysis (Hair, Hollingsworth Carole, Randolph Adriane, & Chong Alain Yee, 2017). The PLS algorithm follows a two-step approach to test the measurement model. The first step aims to calculate the convergent latent variable scores through iteration. The scores are then used to estimate the measurement models (Henseler, Ringle, & Sarstedt, 2012).

As shown in Table 4, the Cronbach's alpha score of each construct was above the recommended value of 0.70. Each construct's composite reliability was above the recommended value of 0.70 (Chin et al., 2003). Every construct's average variance extracted (AVE) was above 0.50, and item loadings on each latent variable were over 0.50, most of which reached higher than 0.7, indicating good convergent validity (Chin et al., 2003).

Table 4
Measurement model.

Construct	Items	Loadings	Cronbach's alpha	Composite reliability	Average variance extracted (AVE)
Individual boundary spanning (BS)	BS1	0.882	0.867	0.910	0.717
	BS2	0.874			
	BS3	0.880			
	BS4	0.743			
Mindfulness (MI)	MI1	0.838	0.898	0.922	0.664
	MI2	0.784			
	MI3	0.852			
	MI4	0.842			
	MI5	0.868			
	MI6	0.691			
Information overload (IO)	IO1	0.888	0.944	0.957	0.816
	IO2	0.915			
	IO3	0.917			
	IO4	0.912			
	IO5	0.882			
Individual work performance (WP)	WP1	0.890	0.916	0.941	0.798
	WP2	0.915			
	WP3	0.880			
	WP4	0.888			
Individual boundary buffering (BB)	BB1	0.827	0.846	0.895	0.741
	BB2	0.801			
	BB3	0.947			

To assess discriminant validity, we compared the square root of each construct's AVE with the absolute value of corresponding correlation coefficients in Table 5. The results indicated that the square roots of all AVEs were greater than their correlations with other constructs, showing satisfactory discriminant validity (Fornell & Larcker, 1981; Gefen & Straub, 2005).

The variance inflation factor (VIF) was calculated, and the values were all below 3.3, indicating that there was no multicollinearity issue (Diamantopoulos & Winklhofer, 2001). We employed Harman's one-factor test to determine whether there was a common method bias issue. The results showed that no factor accounted for over half of the variance, suggesting that common method bias was not a problem (Harman, 1970; Mattila & Enz, 2002).

5.2. Structural model

This study used SmartPLS to calculate latent variable scores of the main model and then used PROCESS macro for SPSS as a supplementary tool to estimate path coefficients. Although SmartPLS can apply the calculated latent variable scores to ordinary least squares regressions (OLS) to estimate path coefficients of the research model without the help of other tools (Henseler et al., 2012), the PLS tool we used (i.e., SmartPLS 2.0M3) could not help us directly obtain specific results of indirect effects of individual boundary spanning on performance, via mindfulness and information overload, respectively. Nor did it enable us to draw the interaction figure between individual boundary spanning and buffering. To deal with these complex model estimation issues, Hayes (2012) developed the PROCESS macro for SPSS, which allowed us to directly obtain detailed path analysis results and draw the two-way

Table 5
The correlation matrix of constructs.

	Mean	Std	1	2	3	4	5
1. BS	4.89	1.26	.847				
2. MI	5.48	0.87	.331**	.815			
3. IO	3.44	1.53	.310**	.074	.903		
4. WP	5.22	1.02	.149*	.320**	-.230**	.893	
5. BB	3.86	1.62	.252**	.124	-.070	.341**	.861

Notes: The bolded elements on diagonal are the square roots of AVEs, and off-diagonal elements are correlations between constructs.

* p < 0.05;

** p < 0.01.

interaction figure. Therefore, this study applied the latent variable scores computed by SmartPLS to PROCESS macro for SPSS to estimate path coefficients. The PROCESS model test that involves the bootstrapping technique can provide a reasonable estimate of indirect effects, and does not require a normally distributed sample (Preacher & Hayes, 2008). This macro can further report detailed mediation results with confidence intervals at the level of 95% as shown in Table 7. SmartPLS and PROCESS are compatible because both use OLS regression to estimate path coefficients (Hair Jr, Hult, Ringle, & Sarstedt, 2016; Preacher & Hayes, 2008). This use of the statistical tool has been applied in prior studies (e.g., Wang et al., 2021; Finoti, Didonet, Toaldo, & Martins, 2017; Gong, 2018).

We tested the main effects (Hypotheses H1a, H1b, H2, and H3) by employing Model 4 of the PROCESS macro and tested the moderating effects by employing Model 7 in PROCESS. To attain significance with a confidence of 95%, the confidence intervals must not include zero. Below, we discuss the results of hypothesis testing.

In Table 6, Model I, Model III, and Model V show the results of the PROCESS Model 4 test; Model II and Model IV show the results of the PROCESS Model 7 test. Fig. 3 shows the results of model testing.

As shown in Table 6, Model I indicates that *individual boundary spanning* is positively related to *mindfulness* ($\beta = 0.294$, $p < 0.01$). Model III shows a positive effect on *information overload* from *individual boundary spanning* ($\beta = 0.252$, $p < 0.01$), and Model V shows that *mindfulness* and *information overload* significantly influence *individual work performance*. ($\beta = 0.271$, $p < 0.01$ and $\beta = -0.291$, $p < 0.01$), thereby supporting Hypotheses 1a and 1b and Hypotheses 2 and 3. We further report the bootstrapping results for the mediation analyses in Table 7. The results reveal that the mediating effects of both *mindfulness* and *information overload* are significant.

Test of Moderating Effects.

H4a and H4b predict that individual boundary buffering moderates the impact of individual boundary spanning on mindfulness and information overload. To test these predictions, the PROCESS formulated the interaction term $BS \times BB$. As shown in Table 6, the interaction term was

Table 6
Regression results.

Variable	Mindfulness		Information overload		Individual work performance Model V
	Model I	Model II	Model III	Model IV	
Polychronicity orientation	0.080	0.074	0.164*	0.196**	-0.081
Non-work ESM use	-0.084	-0.103	0.170*	0.211**	0.055
Age	0.069	0.014	-0.322**	-0.174	0.169
Education	0.278	0.284	-0.190	-0.201	0.014
Gender	-0.014	-0.025	-0.008	-0.037	-0.262
Work experience	-0.100	-0.083	0.058	0.010	-0.052
ESM use experience	0.188**	0.197**	0.219**	0.210**	0.109
Individual boundary Spanning(BS)	0.294**	0.358**	0.252**	0.173*	0.149*
Mindfulness					0.271**
Information overload					-0.291**
Individual boundary Buffering(BB)		-0.021		-0.125	
BS x BB		0.125*		-0.204**	
R ²	0.188	0.205	0.216	0.291	0.238
F	5.567	4.933	6.631	7.837	5.964

Note:

* $P < 0.05$,

** $P < 0.01$.

positively related to *mindfulness* ($\beta = 0.125$ $p < 0.05$) in Model II and negatively related to *information overload* ($\beta = -0.204$ $p < 0.01$) in Model IV. The relevant interaction plots are shown in Fig. 4.

6. Discussion and implications

The advantages and potential negative effects of ESM technologies that facilitate employees' online interactions and access to the enterprise-wide resources pool provide the impetus for this empirical investigation. This study applies an information processing theoretical lens to explore how individuals' boundary spanning and boundary buffering can collectively affect work performance in the ESM context (Dey & Ganesh, 2017). The results of this study support our proposed research hypotheses, which confirmed two distinct mechanisms (i.e., mindfulness and information overload) that could explain the positive and negative effects of ESM-based individual boundary spanning activities. Furthermore, the results indicated that individual boundary buffering could strengthen the positive effect of individual boundary spanning on mindfulness and alleviate the positive influence of individual boundary spanning on information overload. This study links to existing research in several ways.

The present study takes the position that perceives ESM as broadening an individual's resource sources within the organization (Ellison et al., 2015; Leonardi et al., 2013), and introduces the concept of boundary spanning from boundary management theory to explicitly capture the behaviors of individuals obtaining external work resources in the ESM context. Our research validates the positive mediating effect of mindfulness on the relationship between individual boundary spanning and work performance, indicating mindfulness is an effective information processing mode for taking advantage of external work-related resources obtained from individual boundary spanning via ESM. The finding is consistent with a prior study stating that employees' mindfulness can help individuals incorporate task-relevant information obtained from e-mail interruptions during work time (Addas & Pinsonneault, 2018). The revealed mechanism (i.e., mindfulness) adds knowledge to extant boundary spanning research, which recognizes that boundary spanning can broaden resource access and promote work performance (Barner-Rasmussen et al., 2014; Levina & Vaast, 2005; Reiche, 2011).

This study found that individual boundary spanning has a positive linear relationship with information overload, indicating that intensive boundary spanning activities constantly consume individuals' information processing resources, aligning with a prior study's notion that boundary spanning is taxing (Wang et al., 2019). It is worth noting that this finding seems inconsistent with one previous study that found an inverted curvilinear relationship between work-related ESM use and information overload (X. Chen & Wei, 2019). Chen and her colleague argued that as employees become familiar with ESM, they can better manage ESM activities; therefore, information overload decreases. However, the validated moderating role of individual boundary buffering provides a perspective to address such inconsistency. Specifically, an individual who is more familiar with ESM will better understand how to prevent overloading oneself with ESM information (i.e., individual boundary buffering). Our study adds knowledge to the understanding of the relationship between ESM use and information overload.

Lastly, this study examines the moderating role of individual boundary buffering. Previous literature on other information technologies' use behavior suggested taking measures such as queuing (Wilson, 1995), filtering, and withdrawing (Savolainen, 2007) to protect oneself from receiving too much information, thus enhancing one's information processing efficiency. Nevertheless, scholars have pointed out a lack of empirical studies to statistically investigate the effectiveness of approaches for coping with information overload (Saxena & Lamest, 2018). This study echoes this call and provides empirical evidence to consider the protection mechanisms on the effects of ESM on individual performance.

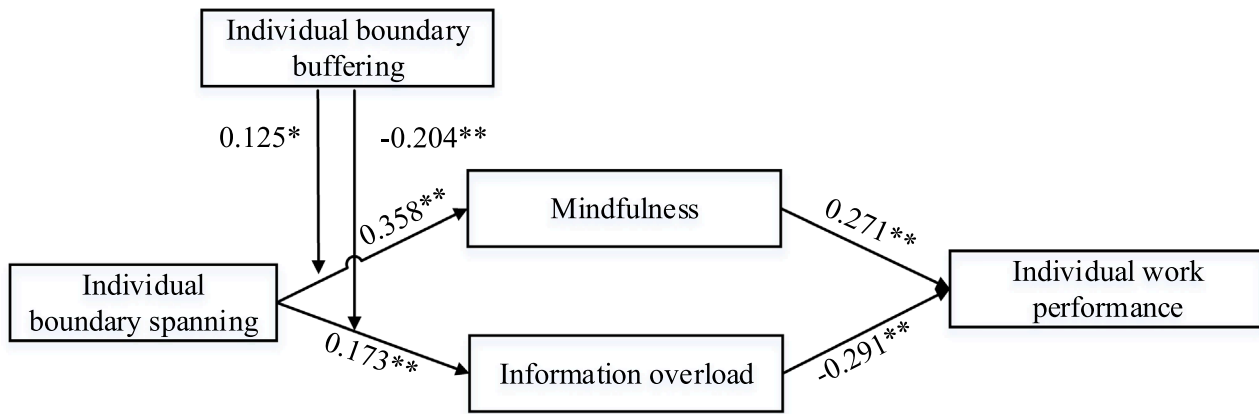


Fig. 3. Results of model testing.

Table 7

Bootstrapping results for the mediation analyses.

Test of mediation	Effect	SE	95% BC CI	
			Lower	Upper
Mindfulness	0.079	0.034	0.027	0.165
Information Overload	-0.073	0.029	-0.140	-0.026

Note: Bootstrapping sample = 5000.

The following section discusses theoretical and managerial implications associated with our findings.

6.1. Research contributions and implications

First, this study contributes to the ESM literature by introducing individual boundary spanning to capture individuals' obtaining external work resource behaviors and exploring mechanisms through which individual boundary spanning can affect individual work performance. Extant studies have validated the fact that ESM's integration of convenient communication and work resource management could facilitate ESM users' obtaining and exchanging work resources (Qi & Chau, 2017; Rode, 2016). For example, Mäntymäki (2016) found that ESM encourages employees to exchange novel work ideas and gather opinions on work issues by reading colleagues' postings on ESM. However, it is a pity that we still know little on whether these resources obtaining and exchanging activities are occurring within the work boundary (e.g., within the project team) or out of the work boundary but within the organization. Such lack of discrimination does not allow us to explicitly capture the advantage of ESM as a tool to broaden individuals' access to

knowledge across their work boundaries as proposed by prior scholars (Ellison et al., 2015; Leonardi et al., 2013). Based on boundary management literature, this study proposed the concept of boundary spanning to explicitly capture individuals' behaviors of obtaining work resources outside the boundary of their work. The study results reveal ESM boundary spanning activities trigger spanners into a mindful processing mode through which they effectively incorporate external work resources obtained to aid their work. These findings echo prior studies' findings that receiving work-related information can benefit one's work (Addas & Pinsonneault, 2018). Simultaneously, due to an individual's limited information processing capacity, intensive boundary spanning could lead to resource deficiency (i.e., information overload) and harm an individual's work performance. This finding is consistent with prior research findings that information overload is a critical negative consequence in ESM use activities (Chen & Wei, 2019; Gao et al., 2018).

Second, the significant moderating effect of ESM boundary buffering suggests that it effectively enhances the value of ESM use. Prior literature presents conflicting views on the relationship between ESM use and work performance. The favorable perspective observed that ESM implementation is an excellent approach to breaking down knowledge silos in organizations and facilitating individuals' sharing and accessing of cross-boundary resources (Ellison et al., 2015; Kwayu, Abubakre, & Lal, 2021; Leonardi et al., 2013). In contrast, the opposing side posited that ESM use depletes individuals' limited resources and negatively influenced them through information overload (Chen & Wei, 2019; Sun et al., 2021). However, limited knowledge exists about alleviating ESM's negative effects on individuals and improving employees' efficiency of resources on ESM (Liang et al., 2020). The results of this study support the prior suggestion of taking measures such as queuing (Wilson, 1995), filtering and withdrawing (Savolainen, 2007) to protect oneself from

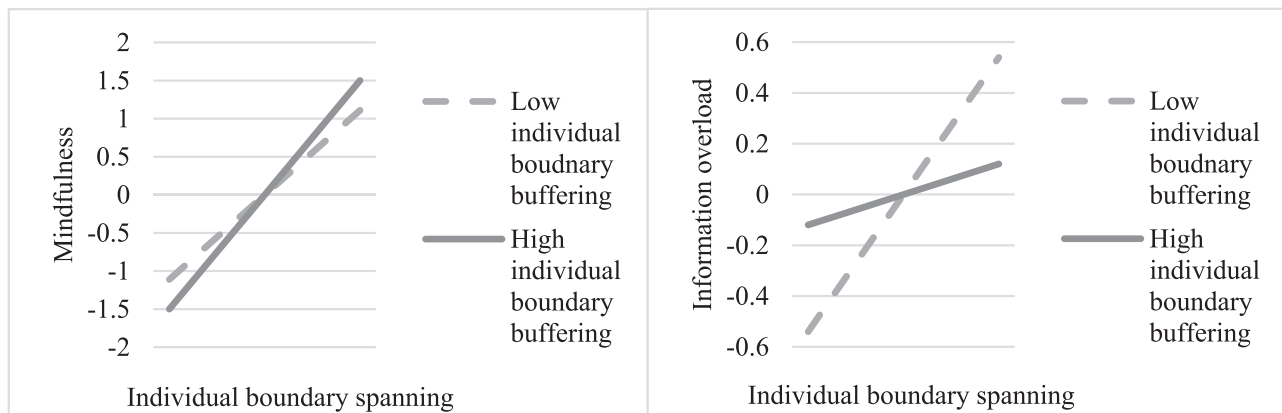


Fig. 4. Moderating effects of individual boundary buffering.

receiving too much information and enhancing one's information processing efficiency. This study provides empirical evidence to consider protection mechanisms in the effects of ESM on individual performance.

Finally, we contribute to the boundary management literature by theoretically proposing and empirically validating that individual boundary spanning and boundary buffering can be strategically combined to effectively benefit an individual's work. Extant studies have found that individual boundary spanning can broaden the channels for acquiring more work resources (Barner-Rasmussen et al., 2014; Levina & Vaast, 2005; Reiche, 2011), facilitating knowledge flow (Pawlowski & Robey, 2004), and providing extra cues for decision-making (Floyd & Wooldridge, 1997), thus benefiting work performance. However, those studies focused solely on the effects of boundary spanning, without in-depth discussion of strengthening the effectiveness of boundary spanning. Our study applied information processing theory and argued that individual boundary spanning via ESM is information-based; such activities can influence individual work performance through two mechanisms—mindfulness and information overload. We further considered the moderating role of individual boundary buffering in the relationship between individual boundary spanning and the two mechanisms. The results revealed that individual boundary buffering activities could strengthen the positive relationship between boundary spanning and mindfulness and alleviate the negative relationship between boundary spanning and information overload, thus, strengthening the effectiveness of individual boundary spanning. Our study echoes Dey and Ganesh's call for studies that do not only consider the influence of separate boundary activities, as this would neglect the interaction effects of boundary activities (Dey & Ganesh, 2017). To the best of our knowledge, this is one of the first few, if not the first, empirical studies on the effects of combining both individual boundary spanning and boundary buffering activities on individual performance.

6.2. Implications for practice

Our study provides significant evidence that employees can enhance work performance by accessing external work resources via ESM. Managers need to realize that employees' boundary spanning activities rely mainly on unofficial communications and interactions. In Chinese culture, co-worker *guanxi* is crucial to interpersonal workplace interactions (Zhang X, 2020), and high-quality co-worker *guanxi* can reduce interaction conflict (Yen, Abosag, Huang, & Nguyen, 2017), facilitate communications, and motivate employees' organizational citizenship behaviors (Lin & Ho, 2010). Therefore, managers can develop a fun and friendly atmosphere in the workplace (Yang, 2018) or organize more joint departmental team-building activities to enhance familiarity among employees, thus strengthening co-worker *guanxi*. On the other hand, team or project leaders can also encourage their members to proactively acquire external work resources via ESM (Faraj & Yan, 2009).

Second, it is notable that the effectiveness of individual boundary spanning also depends on the quality of the external resources. Organizations can take approaches to enrich high-quality work resources documented on ESM (Dang, Zhang, & Chen, 2018), such as providing incentives for employees to encourage ESM content creation and sharing (Beranek, Klein, & Jiang, 2014), setting sharing norms for project teams and departments to require them to upload and maintain the work log and report on the ESM.

Third, although employees can benefit from individual boundary spanning, managers should be aware that employees need to search, recognize work-related information from massive content on ESM, and then incorporate useful information into their work. Such a series of procedures is costly and inefficient. Since individuals possess limited information processing resources, managers are expected to reduce the cost of individuals' boundary spanning activities. Examples include providing a series of content tags, allowing users to customize content tags, and encouraging ESM users to tag the content they create or upload

on the ESM to facilitate other ESM users' content searching. Managers can also establish a reasonable content management directory to better arrange content on the ESM, or increase the communication visibility of ESM to help users develop meta knowledge (i.e., knowledge about who knows what and who knows whom) to assist them in locating the source of needed work resources (Leonardi, 2014).

Fourth, the positive moderating effects of individual boundary buffering show their value during individuals' boundary-spanning activities, especially as ESM functionalities provide various technology-based approaches to support individuals in engaging in these activities. Managers should therefore conduct ESM training courses for employees to clarify and emphasize the use and effects of different functionalities in different typical contexts so that ESM users can have better control over ESM. These practices would ultimately achieve more effective ESM boundary management. The results also imply that the intrusiveness and disturbance of non-work-related activities via ESM during work time are not negligible in the ESM context. Managers should set norms or rules to reduce employees' unnecessary ESM use activities during work time.

Finally, the findings of boundary buffering activities provide directions for ESM developers. They can provide more buffering types of ESM functionalities to improve user experience. For example, by providing tags for information flows to enable ESM users to classify whether the specific information flow is work or non-work-related, such functionality can help ESM manage notifications automatically (e.g., ESM can mute notifications from non-work-related information sources during work time).

6.3. Limitations and future research direction

This study has limitations that present opportunities for future research. First, this study focuses on one typical interaction of boundary management activities. Prior boundary management literature has identified different boundary activities, such as boundary reinforcement (Faraj & Yan, 2009). Future research could follow this direction and seek a new portfolio of boundary management activities. Second, this study considered limited mechanisms regarding the interaction effects of boundary spanning and boundary buffering in the ESM context. Future researchers could investigate other potential mechanisms, such as job burnout or stress, in this specific context (Voydanoff, 2005). Third, although we measured ESM boundary buffering, we did not further discriminate whether the information that individuals' buffer is work-related, which could neglect some of the potential effects of such boundary behaviors. For example, if one buffers work-related information (though he may feel overloaded at that time), such behavior could negatively influence the relationship between him and his colleagues (e.g., collaboration or mutual trust) and work performance. Future research could therefore continue to explore these behaviors to provide finer-grained insights. Fourth, this study provides insights into the effects of individual boundary spanning. However, according to the summary of boundary spanning in the ESM context, there is a lack of research on what factors influence individuals to engage in ESM boundary spanning. For instance, scholars have confirmed different psychological conditions (i.e., psychological meaningfulness, psychological availability, psychological safety) as indispensable factors that affect work activities in the ESM context (Cai et al., 2018). Future studies could follow this direction and explore the relationships between psychological conditions and individual boundary spanning in the ESM context. Finally, the data in this study was collected from one firm in China. Whether the results and discussions are applicable to other countries remains unverified. Different cultures might affect ESM users' behaviors and interactions. Prior studies have suggested differences between Chinese *guanxi* and Western relationship norms (Yang, 2018; Yen et al., 2017). Future studies can apply our model to other cultural contexts to develop more comprehensive research.

7. Conclusion

This study adopts an information processing theoretical lens to empirically identify two mechanisms through which individual boundary spanning affects work performance. Along with these two mechanisms, we also explored the moderating effects of individual boundary buffering via ESM. Our results found that boundary buffering can enhance the positive effects of individual boundary spanning on mindfulness and reduce the side effects of individual boundary spanning on information overload. Our findings emphasize that necessary boundary buffering helps employees better reap the benefits of boundary spanning.

CRedit authorship contribution statement

Jianru Zhang: Methodology, Validation, Formal analysis, Writing –

original draft. **Xiaosong Wu:** Conceptualization, Methodology. **Ju'e Guo:** Funding acquisition. **Randi Jiang:** Writing – review & editing.

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Appendix A. Measurement Items

Construct and items	Sources
Individual boundary spanning BS1. I use ESM to solicit information and resources from external channels. BS2. I make use of my relationships with external colleagues through ESM to support my work. BS3. I actively use ESM to solicit external work resources. BS4. My work depends upon information and resources actively solicited through ESM, that is, information and resources beyond what comes through official channels.	Faraj and Yan (2009)
Information overload IO1. During work time, I have more information on ESM than I have energy to deal with. IO2. During work time, I have more information on ESM than I have time to deal with. IO3. I am often distracted by the excessive amount of information available to me on ESM. IO4. I find that I am overwhelmed by the amount of information I have to process on a daily basis on ESM. IO5. There is too much information about my coworkers on ESM so I find it a burden to handle.	Zhang et al. (2016).
Mindfulness After processing the work-related information from ESM... MI1: I investigated new issues that emerged in my work. MI2: I came out new ideas about doing my work. MI3: I tended to engage in provoking conversations. MI4: I found myself very curious about issues that I faced in my work. MI5: I tried to apply new approach to my work. MI6: I developed an open-mind about the issues I faced, even things that challenged my core beliefs.	Addas and Pinsonneault (2018);Langer (1989)
Individual work performance – Obtained from supervisor ratings Please rate [team member name] on his/her performance on the project in terms of: WP1. Quantity of work output. WP2. Quality of work output WP3. Accuracy of work. WP4. Completing work tasks on-time.	Venkatesh et al. (2018)
Individual boundary buffering BB1. I take measures (e.g., muting nonwork-related notifications) to ensure that I can work free of ESM interference. BB2. I prevent myself from receiving overloaded information or requests on ESM. BB3. I inform others on ESM that I'm concentrating on my work (e.g., by setting a 'busy' mode on ESM).	Faraj and Yan (2009)

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